Assignment

1) What is Power BI and how does it differ from Excel?

→ Power Bi is business intelligence platform from Microsoft that is used to visualize data and share insights. It Focuses on enabling users to connect to hundreds of data sources(cloud, databases, files, etc.),transform data , create models, and visualize the information for data driven decision making.

→ **Difference**

- **Data Capacity** –
  Excel – Up to 1 million rows
  Power BI – Millions to billion rows
- **Security -**

  Excel – File Password, basic

  Power BI – Row level security

- **Visualization –**

  Excel – Static Chart

  Power BI – Interactive Dashboard

- **Automation-**

  Excel – Manual Refresh

  Power BI – Real Time updates

- **Learning Curve-**

  Excel – Easy , Familiar

  Power BI – More Technical, DAX.

2) Explain the concept of data modeling in Power BI.

→ Data modeling in Power Bi is the process of defining the structure, organization and relationships between different data sets that a business collects from one or more sources. It creates a logical representation of how data is interconnected within power bi with tables and relationships, which is essential for creating accurate reports and visualization.

3) What are the different types of connections available in Power BI?

→ Power Bi offers several data connectivity modes.

- Import mode
- DirectQuery mode
- Live Connection
- Composite mode
- Direct Lake

4) How do you handle data transformation in Power BI?

→ We handle Data Transformation in Power BI primarily through the Power Query Editor. This is an ETL tool built into Power BI desktop that allows you to clean , shape, and refine your data before it is loaded into the data model for analysis.

5) What is DAX (Data Analysis Expressions) and why is it important in Power BI?

→ DAX (Data Analysis Expressions) is a formula language developed by Microsoft for Power BI , Power Pivot that enables users to create custom calculations and queries on tabular data models.

→ DAX is critical because it transforms raw, loaded data into meaningful, calculated business intelligence. Without DAX Powe BI limited to simple sums, counts, and averages of existing columns.

6) Can you explain the difference between calculated columns and measures in Power BI?

→ Calculated column is created by adding a new column to an existing table using a DAX formula . It performs calculations foreach row in the table. whereas A measure is a formula written in DAX that calculates values on demand, usually based on filters or context in visual. It performs aggregate or summary level calculations (like totals, averages,counts).

7) How do you handle relationships between tables in Power BI?

→ Handling relationships between tables in Power Bi involves creating and managing connections that allow you to analyse and visualize data from multiple tables accurately.

→ Relationship creates automatically when we load data if columns have the same names and data types. And we can create manually also.

8) What is the purpose of a Power BI Gateway?

→ Power BI Gateway acts as bridge between on-premises data (your local servers, databases, or Files) and the Power BI cloud service.

→ It allows Power BI to securely access and refresh your local data without uploading it to the cloud manually.

9) How can you schedule data refresh in Power BI Service?

→ To schedule data refresh in power bi service follow below steps.

-- Publish -> dataset Setting -> Scheduled Refresh -> Turn On -> Set frequency & time -> Apply

10) Explain the concept of row-level security in Power BI.

→ Row-Level-Security controls who can see what data in Power BI. It filters data based on user roles or login info ensuring secure and personalized access. It protects sensitive data, Gives user personalized, relevant reports.

11) What is the Power BI Desktop and how does it differ from Power BI Service?

→ Power BI desktop is a free application for data analyst and BI developers to connect various data sources , transform and model data , create complex reports with visuals, and perform advance data analysis. It provides extensive data modeling , DAX calculations, and report authoring capabilities.

→ Whereas Power BI service is cloud based SaaS platform where users publish, share and collaborate on reports created in power bi desktop. It focuses on sharing , dashboard, real-time collaborations, managing access, scheduling data refreshes.

12) Explain the concept of Direct Query in Power BI.

→ DirectQuery in Power BI is a connectivity mode that allows Power BI to query data directly from the source at a time of report interaction , instead of importing and storing a copy of the data .When using DirectQuery, Power Bi stores only the schema of the data source, and every time user interacts with a report visual, Power Bi sends queries in real time to the underlying data sources to fetch the latest data.

13) What are Power BI templates and how are they useful?

→ Power BI templates, specifically report templates, are specialized files with a .pbit extension that serve as reusable blueprints for creating reports in Power BI Desktop. Unlike full Power BI report files (.pbix), which include both the structure and the actual data, templates contain only the foundational elements: report pages, visuals, themes, data models (including relationships and measures), queries, and connection configurations.

- **Consistency:** Enforce a standard layout, metrics/measure logic, and data model across departments, regions, or customers.
- **Speed:** Jump-start new reports—no need to rebuild visuals, DAX, or ETL; just point parameters (e.g., server, database, file path, date range) and refresh.
- **Governance and security:** No data is stored in .PBIT files, reducing risk when sharing; model-level settings like Row-Level Security can be pre-defined.
- **Scalability:** Reuse the same design for many tenants or business units with different data sources.
- **Maintenance:** Centralize improvements—update the template model/measures once and redistribute.

14) How do you handle incremental data refresh in Power BI?

→ Incremental data refresh in Power BI allows only new or changed data to be refreshed instead of reloading the entire dataset. It is set up using Rangestart and RangeEnd parameters on a date column, and configured in the Incremental Refresh settings. This improves performance and reduces refresh time.

15) What is the role of Power Query in Power BI?

→ Power Query is a built-in data connectivity, transformation, and preparation tool within Power BI, often referred to as the ETL (Extract, Transform, Load) engine of the platform. It enables users to import data from a wide variety of sources—such as databases, files (e.g., Excel, CSV), web services, or APIs—clean and reshape it to fit analytical needs, and then load the refined dataset into Power BI for visualization and reporting. Unlike direct data loading, Power Query acts as an intermediary layer, ensuring data is structured, consistent, and optimized before analysis, which is crucial for handling messy or disparate data in real-world scenarios.

16) Explain the difference between calculated columns and calculated tables in Power BI.

→ A calculated column is a new column added to an existing table in your Power BI model. It uses a DAX formula to compute values row-by-row, based on the data in that specific row and any related rows via relationships. These values are calculated once during data refresh or model load and stored in the model, making them static until the next refresh

→ A calculated table, on the other hand, is a brand-new table created entirely using a DAX table expression. It doesn't modify an existing table but generates a separate one, often by summarizing, filtering, or aggregating data from one or more source tables.

17) How do you create custom visuals in Power BI?

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18) What are the best practices for optimizing performance in Power BI?

→ Following is best practice for optimize performance in power bi :

- Reducing data volume using filters and aggregations.
- Using **star schema** instead of flat tables.
- Disabling unnecessary **auto date/time**.
- Using **measures** instead of calculated columns.
- Enabling **incremental refresh** for large datasets.
- Optimizing **DAX formulas** and using **import mode** when possible

19) How can you integrate Power BI with other Microsoft products like Azure and Office 365?

→ Power BI integrates seamlessly with other Microsoft products like Azure and Office 365 to enhance data connectivity, analytics, and collaboration.

- **Integration with Azure:**
- Connect Power BI directly to Azure SQL Database, Azure Synapse Analytics, or Azure Data Lake for real-time data access.
- Use Azure Data Factory for data transformation and load data into Power BI.
- Integrate with Azure Machine Learning to apply predictive analytics models in Power BI reports.
- Store large datasets efficiently using Azure Analysis Services or Power BI Premium.

- **Integration with Office 365:**
- Embed Power BI reports in Microsoft Teams for team collaboration.
- Share and analyze data directly from Excel using Power BI Publisher or the "Analyze in Excel" feature.
- Publish dashboards to SharePoint Online for organization-wide access.
- Use Power Automate (Flow) with Power BI to trigger workflows or notifications based on data events.

20) Explain the concept of aggregations in Power BI.

→ Aggregations in Power BI are pre-calculated summaries of detailed data (such as totals, averages, or counts) that improve report performance. Instead of loading or querying large datasets, Power BI uses the aggregated table to answer queries faster. Aggregations are defined in the Model view, linked to detailed tables, and help optimize large models by reducing the amount of data processed during queries.

21) How do you handle error handling and data quality in Power BI?

→ Error handling and data quality in Power BI are managed mainly in Power Query .

- Use tools like **"Remove Errors"**, **"Replace Errors"**, and **"Keep Errors"** to clean data.
- Apply **data type checks**, **null value handling**, and **conditional columns** to fix issues.
- Use **Power Query steps** to trace and correct errors.
- Regularly validate data sources and use **data profiling** (Column quality, distribution, profile) to ensure accuracy and consistency.

22) What is the purpose of Power BI Embedded and when would you use it?

→ **Power BI Embedded** is a Microsoft Azure service that lets developers embed Power BI reports and dashboards into custom web or mobile applications. It allows users to view interactive analytics without needing a Power BI license.

- **Used when:** You want to provide data insights and reporting features inside your own apps or customer portals without redirecting users to Power BI Service. It's ideal for **ISVs** (Independent Software Vendors) and **organizations building analytics into their products**.